

Modeling: transportation problem

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Practice quiz

There are about 45000 Tesla Model S produced per year, that need batteries. 30000 come from the Gigafactory factory in the Nevada, and 15000 from the Gigafactory in Shanghai. Each battery requires 63kg of lithium. Consequently, the company purchases 2835 tons of lithium per year: 1927 tons from mines in Australia, 605 tons from Chili and 303 tons from China.

The transportation costs from the mining sites to the factories are reported in the following table:

Lithium (transp. cost in \$ per ton)			
Factories	Mines		
	Australia	Chile	China
Nevada	8	6	7
Shanghai	4	12	1

How should the transportation be organized to supply the factories at minimum cost? Model the problem as a transshipment problem, and draw the corresponding network. Find the solution using the simplex algorithm.